

Subject Description Form

Subject Code	EIE4121
Subject Title	Machine Learning in Cyber-security
Credit Value	3
Level	4
Pre-requisite	Nil
Co-requisite/ Exclusion	Nil
Objectives	1. To introduce concepts about machine learning techniques in cyber-security 2. To develop skills of using recent techniques for solving practical problems in cyber-security
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand different machine learning techniques 2. Use different techniques for solving problems in cyber security <u>Category B: Attributes for all-roundedness</u> 3. Present ideas and findings effectively
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Machine learning techniques</u> Introduction to machine learning; Basic concepts and classification; Supervised learning and unsupervised learning; classification; clustering; Neural Networks; Support vector machines; Dimensionality reduction; Deep learning techniques such as CNN, LSTM, autoencoder, GAN and other recent structures. 2. <u>Machine learning development environments</u> Software tools for implementing machine learning techniques; Generalization performance; Issues of over-fitting and under-fitting. 3. <u>Malware Analysis</u> Introduction to malware analysis; Types of malware analysis; static analysis, dynamic analysis; Behavioral vs code analysis; Use of different machine learning techniques for malware detection. 4. <u>Phishing detection</u> Introduction to phishing detection; Analysis of email/websites/message features for phishing characterization; Use of different techniques for phishing detection. 5. <u>Anomaly Detection</u> Introduction to the anomaly definition; overview of anomaly detection techniques; static rules technique; use of different machine learning techniques

	for anomaly detection.					
	Laboratory Experiments:					
	Practical Works:					
	1. Introduction to machine learning framework					
	2. Study of feature extraction and evaluation of machine learning techniques in phishing detection					
	3. Study of deep learning techniques in phishing detection					
Teaching/Learning Methodology	Teaching and Learning Method	Intended Subject Learning Outcome	Remarks			
	Lectures	1, 2	Fundamental principles and key concepts of the subject are delivered to students.			
	Tutorials	1, 2	Supplementary to lectures; Students will be able to clarify concepts and to have a deeper understanding of the lecture material; Problems and application examples are given and discussed.			
	Laboratory sessions	2, 3	Students will evaluate different kinds of machine learning techniques.			
	Mini-project	1, 2, 3	Students are required to study the use of machine learning techniques in cyber-security application. Students will need to submit a written report and make a presentation.			
Assessment Methods in Alignment with Intended Learning Outcomes	Specific Assessment Methods/Tasks		% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)		
				1	2	3
	1. Continuous Assessment (total 50%)					
	• Tests		18%	√	√	
	• Laboratory sessions		13%		√	√
	• Mini-project		19%		√	√
	2. Examination		50%	√	√	
	Total		100%			
	The continuous assessment consists of tests, laboratory exercises and a mini-project.					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:					

	Specific Assessment Methods/Tasks	Remark
	Tests	These can measure students' understanding of the theories and concepts as well as their comprehension of subject materials.
	Examination	end-of chapter type problems used to evaluate students' ability in applying concepts and skills learnt in the classroom; students need to think critically in order to come with a solution for a problem.
	Laboratory sessions, mini-project	oral examination will be conducted to evaluate student's technical knowledge and communication skills.
Student Study Effort Expected	Class contact (time-tabled):	
	• Lecture	24 Hours
	• Tutorial/Laboratory/Practice Classes	15 Hours
	Other student study effort:	
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination	26 Hours
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	40 Hours
	Total student study effort:	105 Hours
Reading List and References	1. Zangana, Hewa Majeed, Al-Karaki, Jamal, Omar Marwan, "Revolutionizing cybersecurity with deep learning and large language models", Hershey, PA: IGI Global Scientific Publishing 2025 2. Mark Stamp, Corrado Aaron Visaggio, Francesco Mercaldo, Fabio Di Troia, "Artificial Intelligence for Cybersecurity", Springer International Publishing, 2022. 3. Kumar, Ahlad, Chaudhary, Naveen Kumar, Shastri, Apoorva S, Singh, Mangal, Kulkarni, Anand J, "Digital defense: harnessing the power of artificial intelligence for cybersecurity and digital forensics", CRC Press 2025. 4. UI Rehman, Shafiq, "Challenges and solutions for cybersecurity and adversarial machine learning", IGI Global, 2025. 5. Stamp, Mark, Jurecek, Martin, "Machine learning, deep learning and AI for Cybersecurity", Springer Nature Switzerland, 2025. 6. Thomas Tony, Athira P. Vijayaraghavan, Sabu Emmanuel, "Machine learning approaches in cyber security analytics", Springer, 2020. 7. Padmavathi Ganapathi and D. Shanmugapriya, "Handbook of Research on Machine and Deep Learning Application for Cyber security", IGI Global, 2020.	